

Maven™

Created by <Academia de Código_>

What is Maven?

- Tool for building and managing Java projects
- Manages project dependencies and lifecycle
- Compile, test, package and distribute source code
- Project web site and reports

Why should I move from Ant to Maven?

Ant needs to be told exactly what to do and when to do it,
everything has to be explicitly configured

Maven dependency management system is the reason
most people move away from Ant.

Maven uses convention over configuration
and has lots of implicit functionalities

```
joaofaustino — joaofaustino@joaofaustino — ~ — zsh — 113x8
Last login: Mon Nov 12 12:29:45 on ttys003
[~] ~ mvn -version
Apache Maven 3.5.4 (1edded0938998edf8bf061f1ceb3cfdeccf443fe; 2018-06-17T19:33:14+01:00)
Maven home: /usr/local/Cellar/maven/3.5.4/libexec
Java version: 11, vendor: Oracle Corporation, runtime: /Library/Java/JavaVirtualMachines/jdk-11.jdk/Contents/Home
Default locale: en_GB, platform encoding: UTF-8
OS name: "mac os x", version: "10.14.1", arch: "x86_64", family: "mac"
[~] ~
```

Project Object Model

A Maven project is simply a directory containing a pom.xml file

It is an XML file that contains information about the project and configuration details used to build it

All POMs extend from a default Super POM and describe the project:

- name and version
- source code locations
- dependencies
- plugins
- artifact type

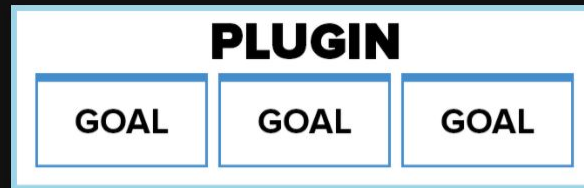
Maven Artifact

An artifact is a file that is produced by the build process and represents an application binary. A Maven artifact can be:

- Binary Package such as a jar, war, ear, zip, tarball, etc.
- Maven Plugin can be of type build or reporting and are executed during the maven lifecycle
- Maven Archetype used as a project template
- Project Metafile, a pom.xml file

Maven Plugins

- Collection of one or more goals
- A goal is a specific task that can be executed



A screenshot of a terminal window. The title bar shows three colored circles (red, yellow, green) and the text "joaofaustino — joaof...". The terminal content shows a green prompt character followed by the command "mvn compiler:testCompile" and a cursor.

Maven Lifecycle

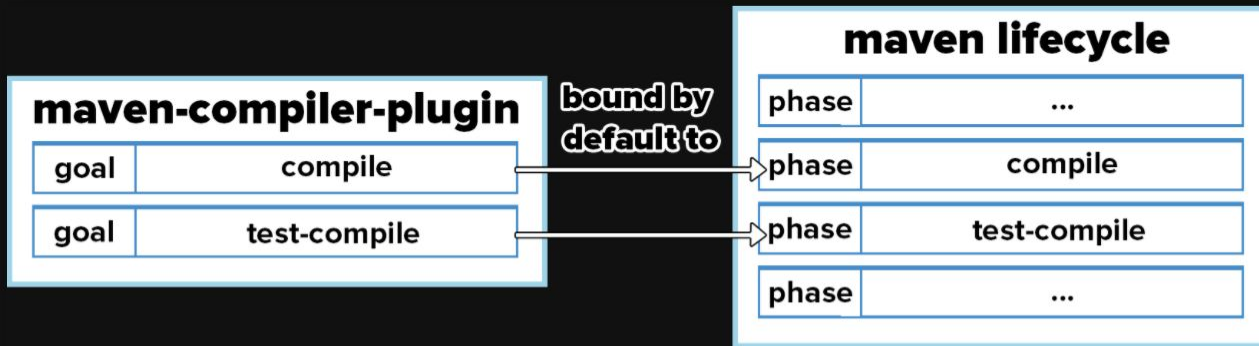
Ordered list of phases which are executed sequentially

DEFAULT PROJECT OBJECT MODEL (POM) BUILD LIFECYCLE

VALIDATE	COMPILE	TEST	PACKAGE	VERIFY	INSTALL	DEPLOY
Validate all necessary information for the project	Compile the source code	Test the compiled source code	Package up compiled source code and other resources	Verify package against criteria	Install the package into the local repository	Deploy the package to a remote repository

```
// Execute all phases up to and including package  
$ mvn package
```

Plugins and Lifecycle



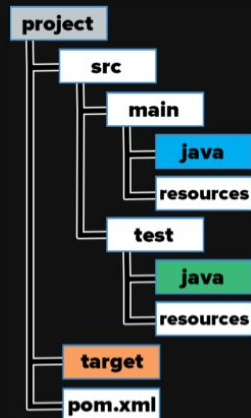
Maven Coordinates

Define a set of identifiers which can be used to uniquely identify a project

- groupId - project group identifier, loosely based on the Java package
- artifactId - name of the project
- version - version of the project in format {major}.{minor}.{patch}, add -SNAPSHOT suffix if in development

```
bootcamp — less exercises/javabank/javabank-step4-tests/pom....
12
13     <dependency>
14         <groupId>org.academiadecodigo.bootcamp</groupId>
15         <artifactId>prompt-view</artifactId>
16         <version>0.2.1-SNAPSHOT</version>
17     </dependency>
18
```

Directory Structure



Maven defines a standard directory structure, allowing users familiar with one Maven project to immediately understand another one

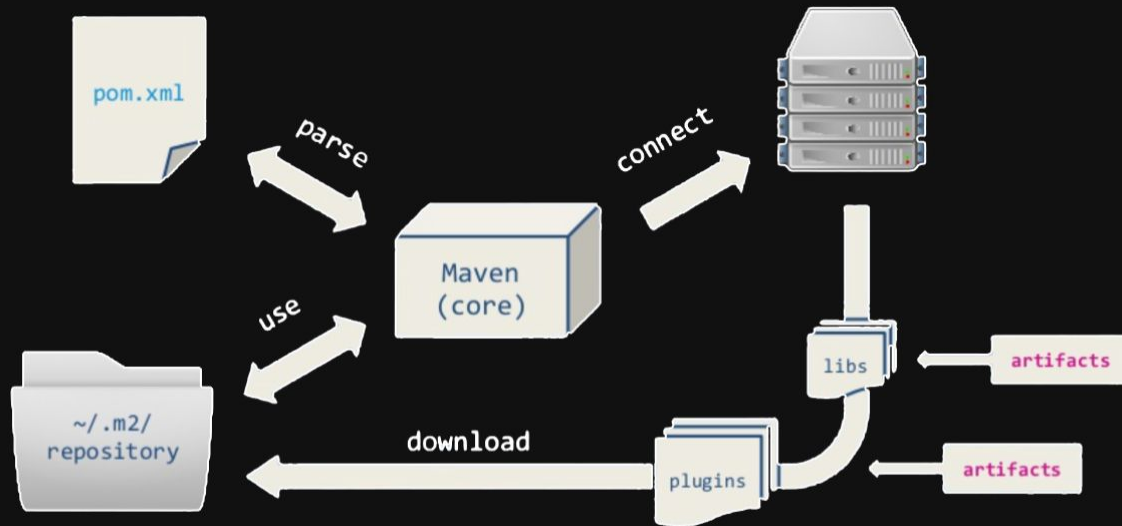
- src/main/java - Application/Library sources
- src/main/resources - Application/Library resources
- src/test/java - Test sources
- src/test/resources - Test resources
- target - output of the build

{ Hands On }

Maven Hello World

Maven Repositories

In Maven, artifacts and plugins are loaded from remote repositories when needed. Downloaded software is then saved in a local repository for use by all maven projects



Dependency Management

We specify in the POM file what external libraries the project depends on, and which version, that Maven will download them into the local repository

If any of these external libraries need other libraries, then these other libraries are also downloaded into the local Maven repository

```
<dependencies>
  <dependency>
    <groupId>com.googlecode.lanterna</groupId>
    <artifactId>lanterna</artifactId>
    <version>2.1.9</version>
    <scope>compile</scope>
  </dependency>
</dependencies>
```

Dependency Scope

Not all dependencies are needed in all situations. Do we really need to package our test code? Dependency scope allows the inclusion of appropriate dependencies for the current stage of the build process

1. compile - default scope and available on the entire lifecycle
2. provided - available on compile and test, provided by the JDK or container at runtime
3. runtime - not required during compile, only on runtime
4. test - compile and run tests
5. system - provided locally, for rare cases
6. import - from another pom

Assembly Plugin

Aggregates the project binaries along with its dependencies, modules, site documentation and other files into a single distributable archive

```
<build>
  <plugins>
    <plugin>
      <artifactId>maven-assembly-plugin </artifactId>
      <groupId>org.apache.maven.plugins </groupId>
      <version>2.6</version>
      <configuration>
        <archive>
          <manifest>
            <mainClass>org.academiadecodigo.LantHelloWorld </mainClass>
          </manifest>
        </archive>
        <descriptorRefs>
          <descriptorRef>jar-with-dependencies </descriptorRef>
        </descriptorRefs>
      </configuration>
      <executions>
        <execution>
          <phase>package</phase>
          <goals>
            <goal>single</goal>
          </goals>
        </execution>
      </executions>
    </plugin>
  </plugins>
</build>
```

{ Hands On }

Maven Jar with Dependencies

Lanterna Hello World